

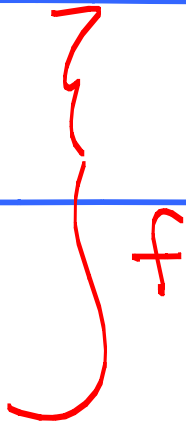
# Eenheid 4.5

Note Title

2015/03/26

Kromme sketsing Stewart p 315

Hoe onthou ek sketsplan?

|   |                                                                                     |            |
|---|-------------------------------------------------------------------------------------|------------|
| D |   | Domain     |
| I |                                                                                     | Intercept  |
| S |                                                                                     | Symmetry   |
| A |                                                                                     | Asymptotes |
| D |  | Decrease   |
| A |                                                                                     | Extremes   |
| C |  | Concavity  |
| I |                                                                                     | Inflection |

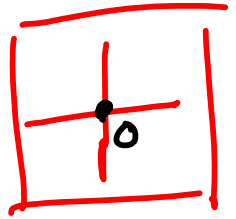
# Voorbeeld

A  $f(x) = x e^{-3x}$

①  $D_f = \mathbb{R}$

② Af-snitte : Y-as :  $x=0 \Rightarrow y=0$

X-as :  $y=0 \Rightarrow x e^{-3x} = 0 \Rightarrow x=0$



③ Symmetrie : • f even | • oneven ( $f(x) = x e^{-3x}$ )

• f even :  $f(-x) = f(x)$

• f oneven :  $f(-x) = -f(x)$

$f(-x) = (-x) e^{-3(-x)} = -x e^{3x} \neq f(x)$  f is nie  
even nie

$f(-x) \neq -f(x)$  f is nie oneven nie

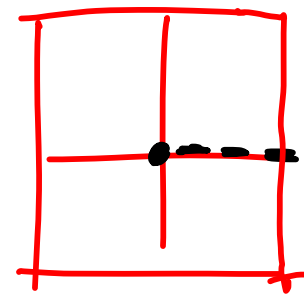
#### ④ • Horizontale asymptote

\*  $x > 0$ :  $\lim_{x \rightarrow \infty} x e^{-3x} (\infty \times 0) = \lim_{x \rightarrow \infty} \frac{x}{e^{3x}} \left( \frac{\infty}{\infty} \right) \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{1}{3e^{3x}} = 0$

Dus HFA is  $y = 0$  as  $x > 0$

\*  $x < 0$ :  $\lim_{x \rightarrow -\infty} x e^{-3x} = \lim_{x \rightarrow -\infty} \frac{x}{e^{3x}} \left( \frac{-\infty}{0} \right) = -\infty$

Geen HFA as  $x < 0$



Vertikale asymptote:  $D_f = \mathbb{R}$   
 $f(x) = x e^{-3x}$  het geen VA nie

#### ⑤ f stygend / dalend

$$f'(x) = \cancel{1} x e^{-3x} + (x)(-3 \cancel{e^{-3x}}) = \cancel{e^{-3x}}_0 (1 - 3x)$$

kritieke punte:  $f'(x) = 0 \Rightarrow e^{-3x} (1 - 3x) = 0 \Rightarrow x = \frac{1}{3}$

$f'$  bestaan oral

Teken van  $f'$

$f'$   $\begin{array}{ccc} ? & & ? \\ & > 0 & < 0 \end{array}$

$\frac{1}{3}$

$f$  stijgt op  $(-\infty, \frac{1}{3})$

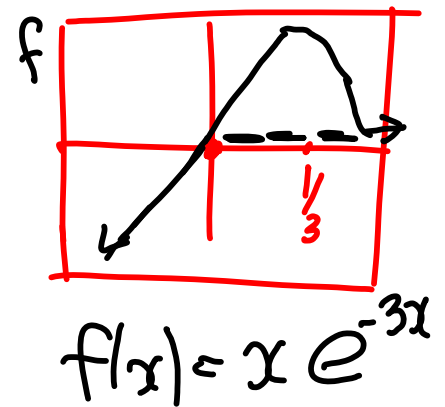
$f$  daalt op  $(\frac{1}{3}, \infty)$

⑥ Lokale extreme

$f$  het in lokale maks van

$$f\left(\frac{1}{3}\right) = \frac{1}{3} e^{(-3)\left(\frac{1}{3}\right)} = \frac{1}{3} e^{-1} = \frac{1}{3e}$$

$$\begin{aligned} f'(x) &= e^{-3x} (1 - 3x) \\ f'(1) &= (+)(-) \\ f'(0) &= (+)(+) \end{aligned}$$



⑦ Konkaviteit:

$$f'(x) = e^{-3x} (1 - 3x)$$

$$f''(x) = -3e^{-3x} (1 - 3x) + e^{-3x} (-3)$$

$$= -3e^{-3x} (1 - 3x + 1) = -3e^{-3x} (2 - 3x)$$

$$f''(x) = 0 \Rightarrow -3e^{-3x} (2 - 3x) = 0$$

$$\Rightarrow x = \frac{2}{3}$$

$$f''(x) = -3e^{-3x} (2 - 3x)$$

$$f''(1) = (-)(-) = (+)$$

$$f''(0) = (-)(+) = (-)$$

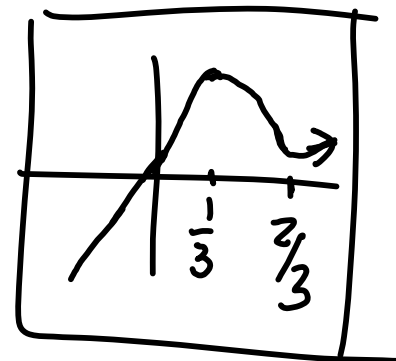
Teken van  $f''(x)$

$< 0 \quad ? \quad > 0$

$\frac{2}{3}$

$f$  is konkav na onder  $\varphi \left(-\infty, \frac{2}{3}\right)$

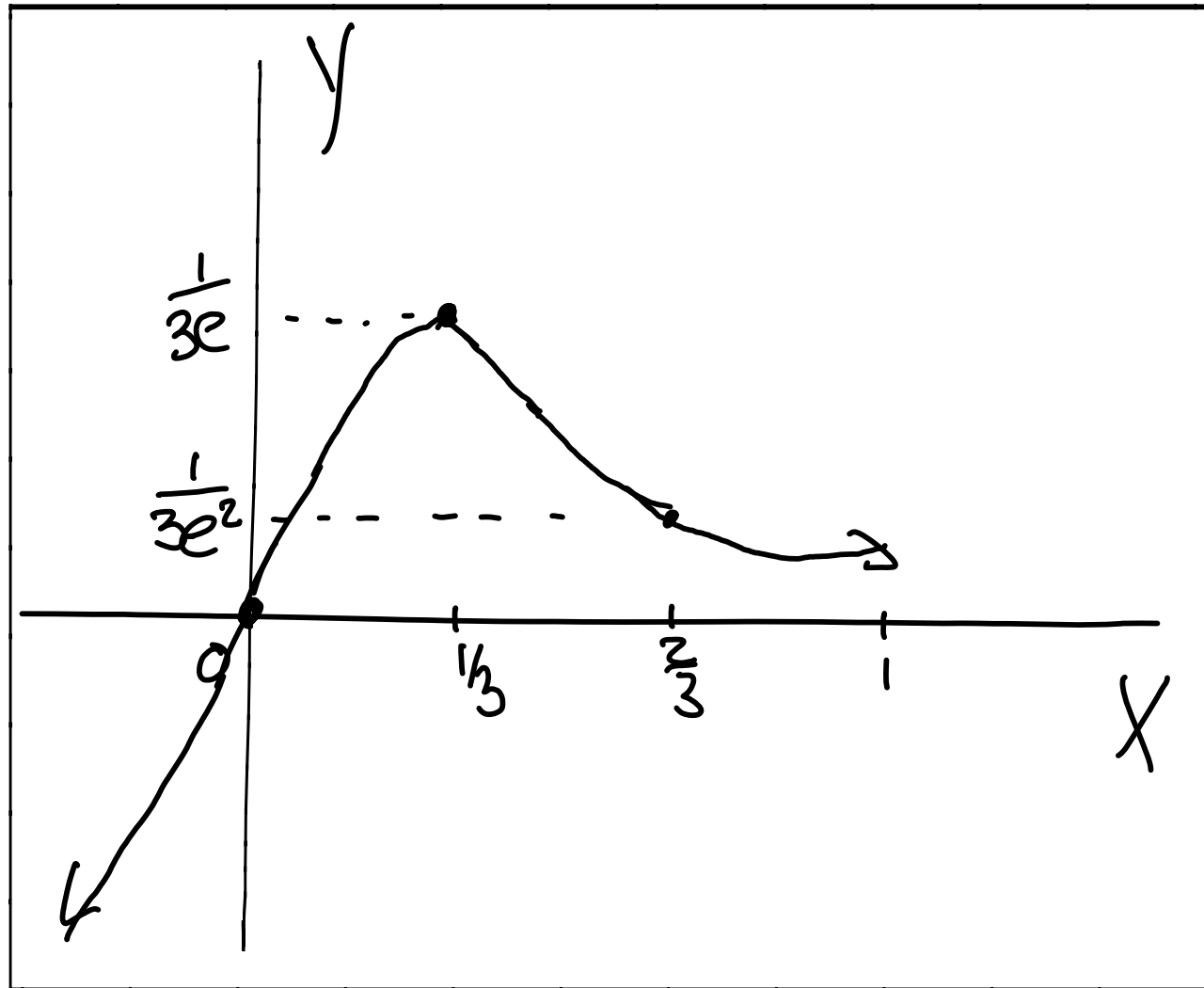
$f$  is konkav na bo  $\varphi \left(\frac{2}{3}, \infty\right)$



④ Inflexiepoint :  $(x, f(x))$   $f(x) = x e^{-3x}$   
 $x = \frac{2}{3}$   $f(\frac{2}{3}) = \frac{2}{3} e^{-3(\frac{2}{3})} = \frac{2}{3} e^{-2} = \frac{2}{3e^2}$

Sketch:

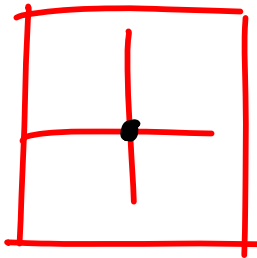
$$f(x) = x e^{-3x}$$



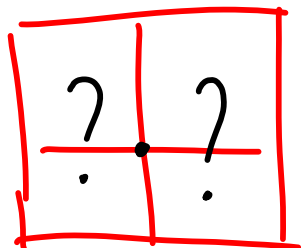
B  $f(x) = \ln(1+x^2)$

①  $D_f = \mathbb{R}$

② Afssnitte:  $y$ -as:  $x=0 \Rightarrow y = \ln 1 = 0$   
 $x$ -as:  $y=0 \Rightarrow e^{\ln(1+x^2)} = e^0$   
 $\Rightarrow 1+x^2 = 1 \Rightarrow x^2 = 0 \Rightarrow x=0$

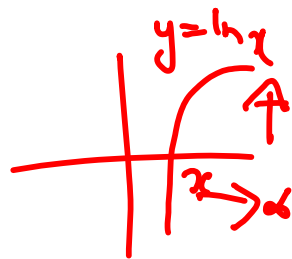


③ Symmetrie: f even oder ungerade?  
 $f(-x) = \ln(1+(-x)^2) = \ln(1+x^2) = f(x)$   
 $\therefore f$  is even



④ Horizontale asymptote:

$x > 0: \lim_{x \rightarrow \infty} \ln(1+x^2) = \infty$



$$x < 0 : \lim_{x \rightarrow -\infty} \ln(1+x^2) = \infty \quad \text{Geen HF}$$

Vertikale asymptoot:  $D_f = \mathbb{R}$

Geen vA

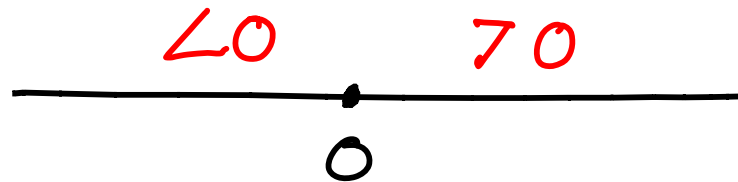
⑤ f stijgend | dalend :  $f(x) = \ln(1+x^2)$

$$f'(x) = 0 \Rightarrow f'(x) = \frac{2x}{1+x^2} = 0 \Rightarrow 2x = 0 \Rightarrow \underline{\underline{x=0}} \in D_f$$

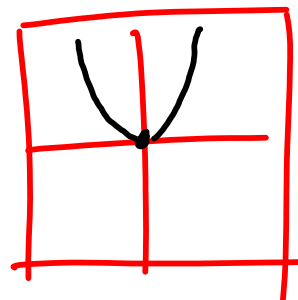
$f'(x)$  bestaat al  $\therefore$  kritieke punt  $c=0$

Teken van

$f'$



\*  $f$  stijgt op  $(0, \infty)$   
 $f$  daalt op  $(-\infty, 0)$



$$f'(x) = \frac{2x}{(1+x^2)}$$

$$f'(1) = \frac{+}{+}$$

$$f'(-1) = \frac{-}{+}$$



⑥ Lokale ekstreme:  
 Lokale min / maks ?? (y-waarde)  
 Lokale min is  $f(0) = \ln(1+0) = 0$

⑦ Konkaviteit  $f'(x) = \frac{2x}{1+x^2}$

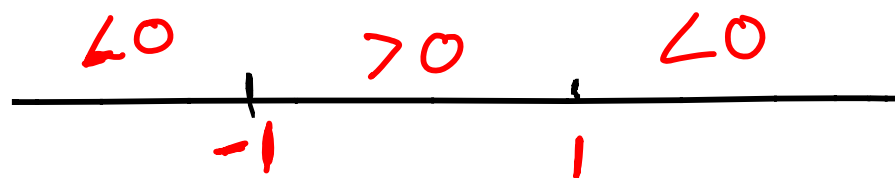
$$f''(x) = \frac{(2)(1+x^2) - (2x)(2x)}{(1+x^2)^2}$$

$$= \frac{2 + 2x^2 - 4x^2}{(1+x^2)^2} = \frac{-2x^2 + 2}{(1+x^2)^2} = \frac{-2(x^2 - 1)}{(1+x^2)^2}$$

$$= \frac{-2(\underline{x-1})(\underline{x+1})}{(1+x^2)^2}$$

•  $f''(x) = 0 \Rightarrow x = 1$  of  $x = -1$

Teken van  
 $f''$



|                                                                   |                        |
|-------------------------------------------------------------------|------------------------|
| $f''(x) = \frac{-2(\underline{x-1})(\underline{x+1})}{(1+x^2)^2}$ |                        |
| $f''(-2) = \frac{(-)(-)(-)}{+}$                                   | $f''(2) = \frac{-}{+}$ |
| $f''(0) = \frac{(-)(-)(+)}{+}$                                    |                        |

$f$  is konkkaaf na bo  $q$   $(-1, 1)$   
 $f$  is konkkaaf na onder  $p$   $(-1, -1)$  en  $q$   $(1, 1)$

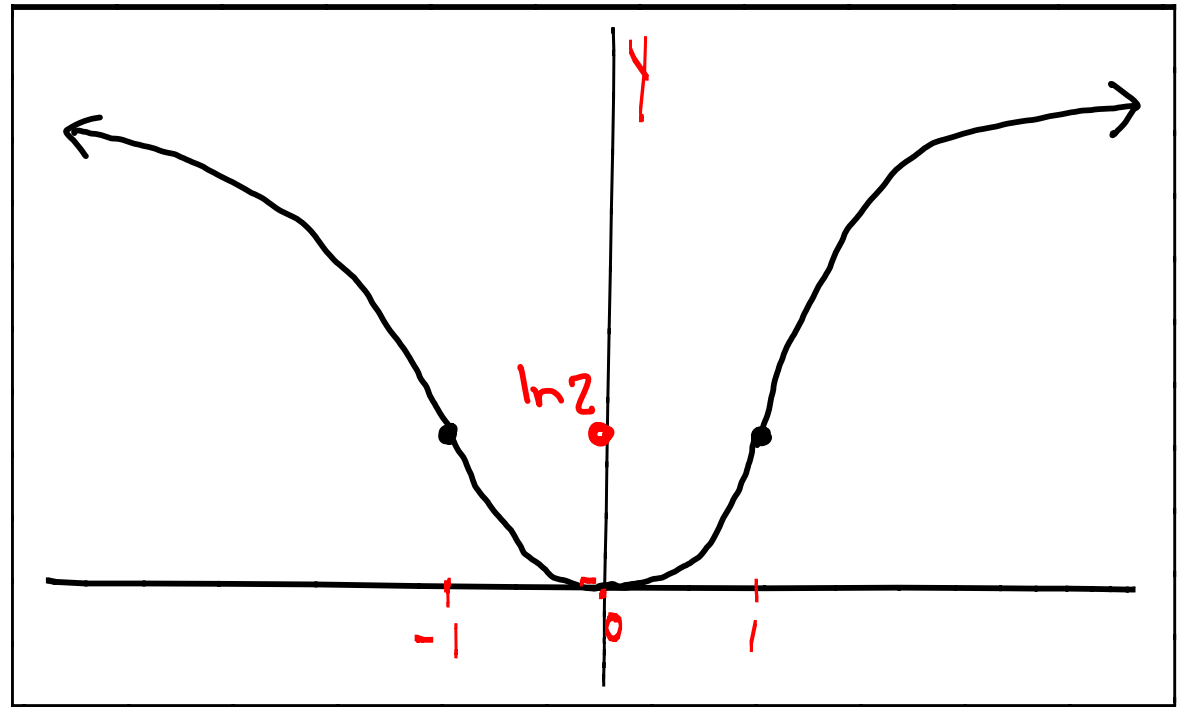
⑧ Inflexiepunten:  $(-1, f(-1))$  en  $(1, f(1))$   
 $\therefore (-1, \ln 2)$  en  $(1, \ln 2)$

Skets

$$f(x) = \ln(1+x^2)$$

$$f(1) = \ln 2$$

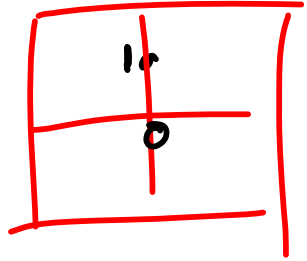
$$f(-1) = \ln 2$$



C  $f(x) = \frac{e^{2x}}{x+1} \quad (x \neq -1)$

①  $D_f = (-\infty, -1) \cup (-1, \infty) = \mathbb{R} \setminus \{-1\}$

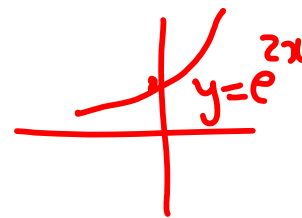
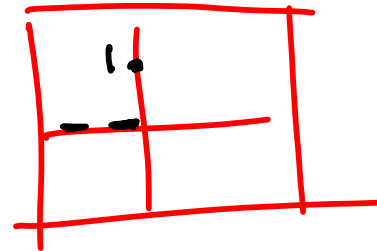
② afsnitte: y-as :  $x=0 \Rightarrow y=1$   
 x-as :  $y=0$  Geen afsnit



③ Symmetrie: f ewe | onewe  
 f Nie ewe | onewe

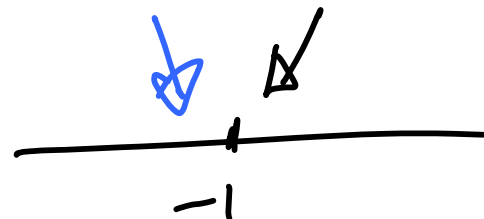
④ Horizontale asymptoot  
 $x > 0$  :  $\lim_{x \rightarrow \infty} \frac{e^{2x}}{x+1} \left( \frac{\infty}{\infty} \right) \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{2e^{2x}}{1} = \infty$  Geen HA as  $x > 0$

•  $x < 0$  :  $\lim_{x \rightarrow -\infty} \frac{e^{2x}}{x+1} \stackrel{0}{=} 0$   
 $\therefore$  HA is  $y = 0$



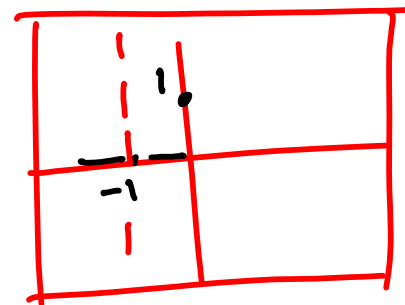
• Vertikale asymptote (doen albei kante) van  $-1$

$\lim_{x \rightarrow -1^+} \frac{e^{2x}}{1+x} = +\infty$



$\lim_{x \rightarrow -1^-} \frac{e^{2x}}{1+x} = -\infty$

$\therefore$  VA is  $x = -1$



⑤ f stijgend / dalend :

$$f'(x) = \frac{(2e^{2x})(x+1) - e^{2x}}{(x+1)^2}$$

$$= \frac{e^{2x}(2x+2-1)}{(x+1)^2} = \frac{e^{2x}(2x+1)}{(x+1)^2}$$

- $f'(x) = 0 \Rightarrow \underbrace{e^{2x}}_{>0} (2x+1) = 0 \Rightarrow 2x+1 = 0$   
 $x = -\frac{1}{2} \in D_f$

- $f'$  Bestaan nie as  $x = -1$  Maar  $x = -1 \notin D_f$

kritieke getalle: (Enigste)  $c = -\frac{1}{2}$

Teken van  $f'$  (uitgesluit  $VP$ )

$< 0$      $< 0$      $> 0$   
 $\quad \quad \quad -1 \quad \quad -\frac{1}{2}$

$f$  daal op  $(-\infty, -1)$  en op  $(-1, -\frac{1}{2})$

$f$  styg op  $(-\frac{1}{2}, \infty)$

|                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------|
| $f'(x) = \frac{e^{2x}(2x+1)}{(x+1)^2}$ $f'(-2) = \frac{(+)(-)}{(+)}$ $f'(-\frac{3}{2}) = \frac{(+)(-)}{(+)}$ $f'(0) = \frac{(+)(+)}{(+)}$ |
|-------------------------------------------------------------------------------------------------------------------------------------------|

⑥ Lokale ekstreme:

Lokale min  $f(-\frac{1}{2}) = \frac{e^{2(-\frac{1}{2})}}{\frac{1}{2}} = \frac{2}{e}$

⑦ Konkavität:  $f'(x) = \frac{e^{2x}(2x+1)}{(x+1)^2}$

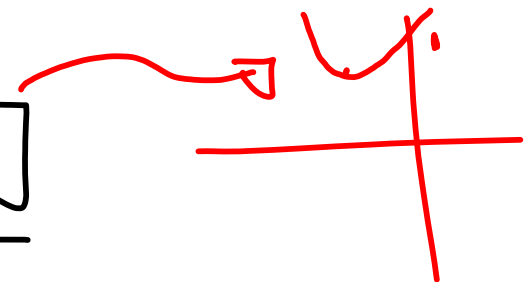
$$f''(x) = \frac{[2e^{2x}(2x+1) + 2e^{2x}](x+1)^2 - [e^{2x}(2x+1)](2)(x+1)}{(x+1)^4}$$

$$= \frac{(2e^{2x})[(2x+1)+1](x+1)^2 - (2e^{2x})(2x+1)(x+1)}{(x+1)^4}$$

$$= \frac{(2e^{2x})(x+1)[(2x+2)(x+1) - (2x+1)]}{(x+1)^4}$$

$$= \frac{(2e^{2x})(x+1)[2x^2 + 4x + 2 - 2x - 1]}{(x+1)^4}$$

$$= \frac{(2e^{2x})\cancel{(x+1)}[2x^2 + 2x + 1]}{(x+1)^3}$$



Teken van  $f''$

neg  $\textcircled{+}$  pos

$-1$

$f$  is konkav na bo  $op$   
 $(-1, 0)$

$f$  is konkav na ander  $op$   $(-\infty, -1)$

$$f''(x) = \frac{(2e^{2x})(2x^2 + 2x + 1)}{(x+1)^3}$$

$$f''(-2) = (+)(+)$$

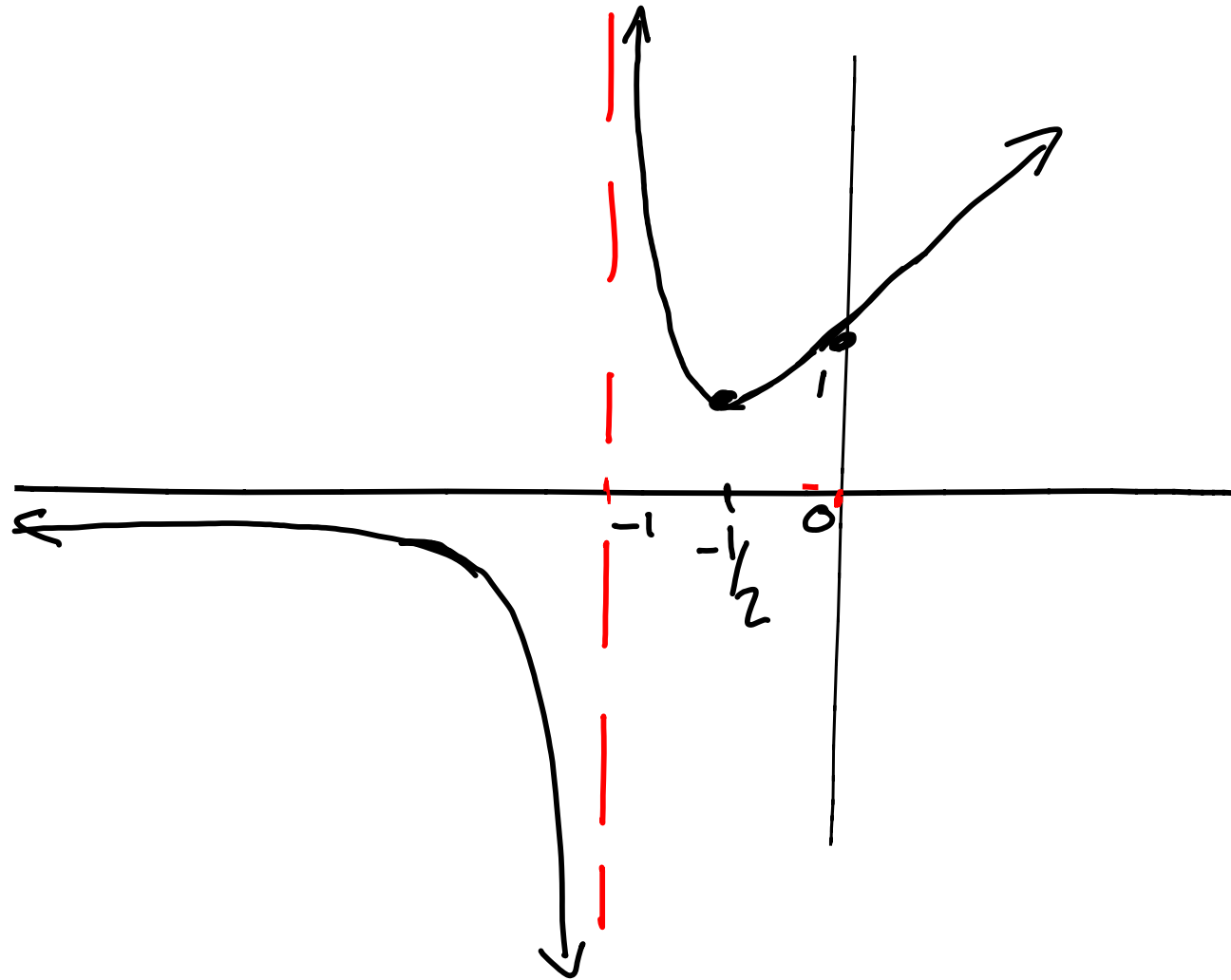
$$f''(0) = \frac{(+)(+)}{+}$$

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Inflexie punt :

Geen : VA is  
 $x = -1$

Sketch:  $f(x) = \frac{e^{2x}}{x+1}$





# \*Eenheid 4.6      Optimering

Nie vir eksamendoeleindes nie

Bv  
↳ Wildsbaer het draad vir 'n wildsheining van 1200 m om 'n nuwe wildskamp beskikbaar. Aan die een kant is 'n rivier.

Bepaal die maksimum opp.  
wat hy kan toespans.



Oplossing: Maksimum Opp / Area

$$Opp = xy$$

$$Omtrek = y + 2x = 1200$$

$$\Rightarrow y = 1200 - 2x \quad \textcircled{1}$$

Vervang  $\textcircled{1}$  in opp

$$\begin{aligned} Opp &= x(1200 - 2x) = A(x) \\ &= 1200x - 2x^2 \end{aligned}$$

Maksimum opp:

$$\begin{aligned} 1200x - 2x^2 &= 0 \\ \Rightarrow x(1200 - 2x) &= 0 \\ \Rightarrow x = 0 \text{ of } x = 600 \end{aligned}$$

$$\therefore 0 \leq x \leq 600$$

Gesloten interval  
methode

kritieke getal:  $\bullet F'(x) = 1200 - 4x = 0$   
 $\Rightarrow x = 300 \in D_f = [0, 600]$

$\bullet F'(x)$  bestaat wel

|     |   |         |     |
|-----|---|---------|-----|
| $x$ | 0 | 300     | 600 |
| $y$ | 0 | 180 000 | 0   |

Absolute maks  
by  $x = 300$   
180 000 m<sup>2</sup>